

题目

某有机溶剂可以从水中萃取水杨酸，设其分配系数 $K_D = 100$ ，水杨酸二聚平衡常数 $K = 50$ 。已知水杨酸的 $pK_{a1} = 2.98$, $pK_{a2} = 13.66$ 。

有以下问题：

- 计算 $pH = 1.0$ 时，原水中水杨酸总浓度为 0.1mol/L 时等体积的该溶剂在水中萃取水杨酸的分配比 D 。（答案取三位有效数字）
- 计算人体生理条件下，原水中水杨酸总浓度为 0.1mol/L 时等体积的该溶剂在水中萃取水杨酸的分配比 D 。（答案取两位有效数字）
- 计算分配比 $D = 1$ 时水体系的 pH 。（答案取一位有效数字）

下列说法正确的是（计算误差在2%内可视为正确）：

- 其他选项均不正确
- 问题a的答案大于400
- 问题a的答案小于360
- 问题b的答案为0.0021
- 问题b的答案为0.0051
- 问题c的答案为5.2
- 问题c的答案为5.6

H. 问题c的答案为5.4

答案

正确答案: **H**

详细解析

根据水杨酸的二级电离常数可知，水杨酸二级电离极其微弱，可忽略。因此将其记作 **HA**。

CHECKPOINT

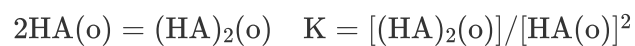
1 PTS

水杨酸二级电离极其微弱，可忽略

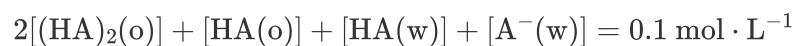
后缀 (w) 表示水中物种，(o) 表示有机溶剂中物种。

问题a:

根据平衡式有：



等体积萃取, 故物料守恒可写作:



CHECKPOINT

1 PTS

物料守恒 $2[(\text{HA})_2(\text{o})] + [\text{HA}(\text{o})] + [\text{HA}(\text{w})] + [\text{A}^-(\text{w})] = 0.1 \text{ mol} \cdot \text{L}^{-1}$

pH = 1 时,有 $\delta_{\text{HA}} \approx 1$, 忽略 $[\text{A}^-(\text{w})]$

CHECKPOINT

1 PTS

pH = 1 时,有 $\delta_{\text{HA}} \approx 1$, 忽略 $[\text{A}^-(\text{w})]$

整理代入得:

$$2K[\text{HA}(\text{o})]^2 + (\text{K}_{\text{D}} + 1)[\text{HA}(\text{o})]/\text{K}_{\text{D}} - 0.1 = 0$$

解得

$$[\text{HA}(\text{o})] = 0.0270 \text{ mol} \cdot \text{L}^{-1}$$

$$[\text{HA}(\text{w})] = 0.000270 \text{ mol} \cdot \text{L}^{-1}$$

$$[(\text{HA})_2(\text{o})] = 0.0364 \text{ mol} \cdot \text{L}^{-1}$$

$$\text{D} = (2[(\text{HA})_2(\text{o})] + [\text{HA}(\text{o})])/[\text{HA}(\text{w})] = 370$$

CHECKPOINT

1 PTS

$$D = (2[(\text{HA})_2(\text{o})] + [\text{HA}(\text{o})])/[\text{HA}(\text{w})] = 370$$

问题b:

人体生理环境的 $\text{pH} = 7.4$, 此时有 $\delta_{\text{A}^-} \approx 1$, 忽略 $[\text{HA}(\text{w})]$ 。

CHECKPOINT

1 PTS

人体生理环境的 $\text{pH} = 7.4$, 此时有 $\delta_{\text{A}^-} \approx 1$, 忽略 $[\text{HA}(\text{w})]$ 。

因此有:

$$[\text{HA}(\text{o})]/[\text{A}^-(\text{w})] = K_{\text{D}}[\text{H}^+]/K_{\text{a1}} = 0.00380$$

CHECKPOINT

1 PTS

$$[\text{HA}(\text{o})]/[\text{A}^-(\text{w})] = K_{\text{D}}[\text{H}^+]/K_{\text{a1}} = 0.00380$$

带入物料守恒整理可得

$$100[\text{HA}(\text{o})]^2 + 264[\text{HA}(\text{o})] - 0.1 = 0$$

解得

$$[\text{HA}(\text{o})] = 0.000379 \text{ mol} \cdot \text{L}^{-1}$$

$$[\text{A}^-(\text{w})] = 0.10 \text{ mol} \cdot \text{L}^{-1}$$

$$[(\text{HA})_2(\text{o})] = 0.0000144 \text{ mol} \cdot \text{L}^{-1}$$

$$D = (2[(\text{HA})_2(\text{o})] + [\text{HA}(\text{o})]) / [\text{A}^-(\text{w})] = 0.0041$$

CHECKPOINT

1 PTS

$$D = (2[(\text{HA})_2(\text{o})] + [\text{HA}(\text{o})]) / [\text{A}^-(\text{w})] = 0.0041$$

问题c:

$D = 1$, 显然pH在1与7.4之间, 忽略二级电离。

物料守恒: $2[(\text{HA})_2(\text{o})] + [\text{HA}(\text{o})] = 0.05 \text{ mol} \cdot \text{L}^{-1}$; $[\text{HA}(\text{w})] + [\text{A}^-(\text{w})] = 0.05 \text{ mol} \cdot \text{L}^{-1}$

CHECKPOINT

1 PTS

物料守恒: $2[(\text{HA})_2(\text{o})] + [\text{HA}(\text{o})] = 0.05 \text{ mol} \cdot \text{L}^{-1}$; $[\text{HA}(\text{w})] + [\text{A}^-(\text{w})] = 0.05 \text{ mol} \cdot \text{L}^{-1}$

平衡常数代入整理得:

$$100[\text{HA}(\text{o})]^2 + [\text{HA}(\text{o})] - 0.05 = 0$$

$$[\text{HA}(\text{o})] = 0.0179 \text{ mol} \cdot \text{L}^{-1}$$

$$[(\text{HA})_2(\text{o})] = 0.016 \text{ mol} \cdot \text{L}^{-1}$$

$$[\text{HA}(\text{w})] = 0.000179 \text{ mol} \cdot \text{L}^{-1}$$

$$[\text{A}^-(\text{w})] = 0.05 \text{ mol} \cdot \text{L}^{-1}$$

$$[\text{H}^+] = K_a[\text{HA}(\text{w})]/[\text{A}^-(\text{w})] = 3.75 \times 10^{-6} \text{ mol} \cdot \text{L}^{-1}$$

$$\text{pH} = 5.42$$

CHECKPOINT

1 PTS

$$\text{pH} = 5.42$$

由于pH有效数字计算其小数点后的位数，故保留一位有效数字为5.4。

因此选项H正确。